



Georgia Institute of Technology

Draft Report Deliverable
Bandsaw Monitor

Issued by: Rachel Lohan

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Table of Contents

Table of Contents.....	2
Introduction.....	2
Solution Overview.....	2
System Architecture.....	3
Software Design and Refactor.....	3
Core Functionality.....	4
Evaluation.....	4
Successes.....	4
Pitfalls.....	6
Conclusion and Future Work.....	7
References.....	8

Introduction

The operation of industrial bandsaws in educational makerspaces presents a safety challenge.[2] Users in these environments range from first-time users to experienced users who may develop unsafe habits over time. A common failure is the loss of awareness between the user's hands and the cutting blade, which can result in severe injury. Existing safety solutions like physical guards or intrusive machine-integrated systems can reduce usability. This innovation aimed to increase safety with an easy to-install, unintrusive, and intuitive system.

This project proposes a computer vision based safety system that enhances user awareness without modifying the bandsaw itself. [3] Rather than physically preventing unsafe actions, the system continuously monitors the cutting area and provides immediate feedback when hazardous behavior is detected.

The initial goal of the project was to improve upon an existing system capable of detecting hand proximity to the blade and issuing real-time alerts. Making the system as informative, professional and scalable for future students as possible. Over the course of development, the project effort was mostly directed toward improving software structure and reliability through refactoring, unit testing and documentation.

Solution Overview

The final system consists of a Raspberry Pi 5 as the main processor and two-axis gimbal-mounted camera for hand detection which is packaged into a magnetic mountable device. The large change made this semester was removing the string LED lights that were used for indication to an informative GUI hosted on the existing screen of the raspPI. This change was made to solidify the design component and make the overall user interface more useful.

The system works by capturing video feed of the bandsaw cutting region, detecting the user's hands using a computer vision model, and evaluating wheather those hands intersect the defined danger zone near the blade. When unsafe conditions are detected, the system changes the GUI screen to reflect the behavior. So, the screen is always displaying either the safe indicator or the unsafe indicator if detected.

System Architecture

The system is composed of three primary subsystems:

1. **Sensing and Input**

A camera module continuously captures frames of the cutting region. The positioning of the camera is adjustable via a custom mount

2. **Processing and Detection**

A Raspberry Pi 5 processes [1] incoming frames through the detection pipeline. The pipeline includes:

- frame acquisition
- preprocessing
- hand detection
- danger zone calibration
- intersection and proximity analysis

3. **Output and Alerting**

A GUI provides immediate visual feedback when unsafe conditions are detected. An informative image + color will alert the user.

Software Design and Refactor

A major portion of the project involved restructuring the software system into a more scalable and flexible integration.

We removed global variables, formed classes and removed hardcoded materials. Previous versions of the code made debugging difficult and limited the system's ability. Through refactoring, the system was organized into the following: video capture, detection, state evaluation and control. This separation of concerns significantly improved readability, maintainability, and debugging efficiency.

The addition of Robot tests also made the refactor process smoother through continuous testing functionality being possible.

Core Functionality

The final system is capable of the following:

1. Capture video frames from the camera
2. Detect hands and generate bounding boxes
3. Identify the blade location and define a danger zone
4. Evaluate whether any detected hand intersects or approaches this zone
5. Trigger GUI alerts when unsafe conditions are detected

This pipeline operates continuously during bandsaw use, providing immediate feedback to the user without interrupting machine function.

Evaluation

Successes

One of the most impactful design decisions was changing the alerting mechanism of the project to the GUI over the string lights. This would make the string light component unnecessary meaning less components overall streamlining the entire design. It also meant that we have the freedom to implement more informative indications to the user. For example, prior to this the string light would light either red or green during use to alert the user to unsafe practices. But now, the GUI shows a basic image to the user that denotes the dangerous or safe practice with the colored background displayed as well (red/green). The GUI was made using PyQt and functions when the monitor is turned on.

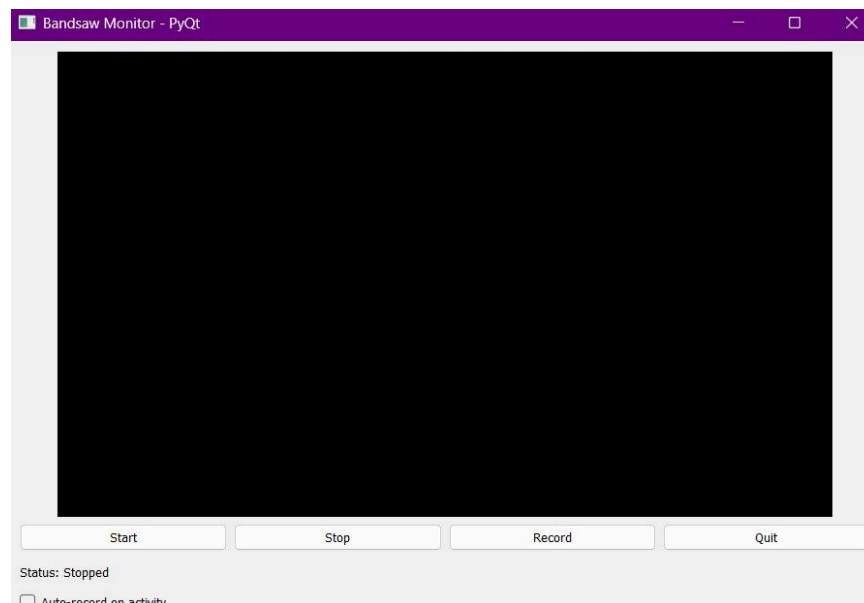


Figure 1: GUI PyQt

Another major success was the software refactor. Early in development, the codebase lacked clear structure, which made it difficult to debug and extend. By reorganizing the system into modular components, development became significantly more efficient. This restructuring also reduced onboarding complexity for new contributors and improved overall system reliability.

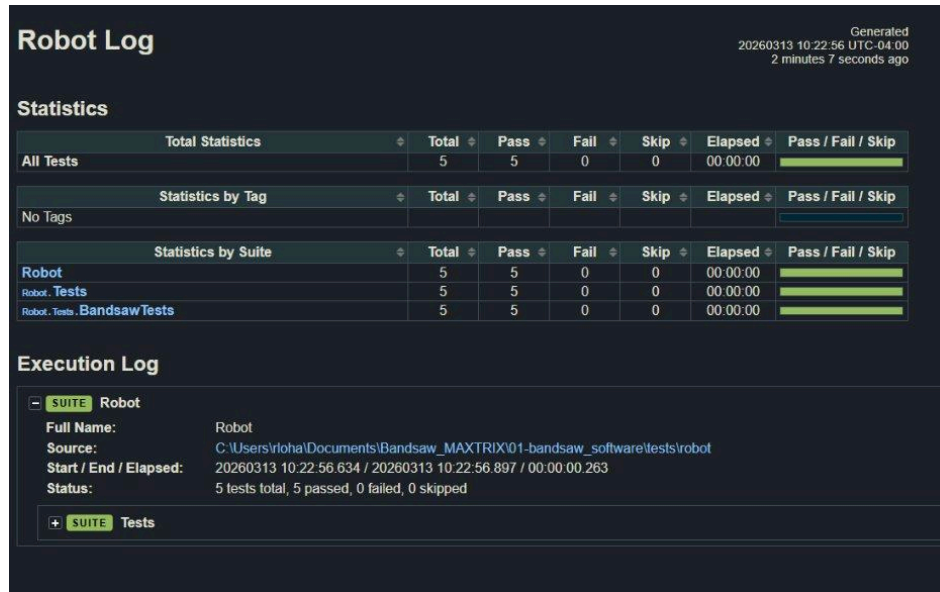


Figure 2: Robot Test cases

To add on to this refactor, Robot system tests were also added to ensure code functionality even with a large refactor. During the refactor, this testing framework provided immediate feedback whenever a change introduced a regression. Instead of relying on manual testing or visual inspection of outputs, we were able to quickly identify and isolate issues within specific modules. This significantly reduced debugging time and allowed for more confident restructuring of the codebase. The introduction of automated testing improves the long-term maintainability of the project. Future development can be validated against the existing test suite to ensure that core system behavior remains intact.

In order to aid developers in the future, a quickstart guide was also set up. This guide includes an introduction to the system, a high level overview, common challenges, setup instructions and further explanation of the codebase and how to use it.

Pitfalls

Some areas that we can highlight for future improvement are final hardware design, GUI validation, and real-testing.

The final hardware design was a pitfall during this project for development. This was due to inexperience with material the bandsaw group wanted to pursue and external factors. This is now seen as a large area for the next round of iterative improvements.

Another pitfall was the initial onboarding to the project. The project did not have a github or code repository and the only setup details were extremely high level and abstract. This was mitigated

with the development and addition of those items that the project was lacking in. The github is now maintained and a quickstart guide spun-up.

Another factor would be getting further feedback on the GUIs functionality and user overtime in the makerspace. This could inform more testing and more improvement on the UI side of the project to enhance usability. A final thought could be to add an auditory alert significant enough to be heard over the machinery.

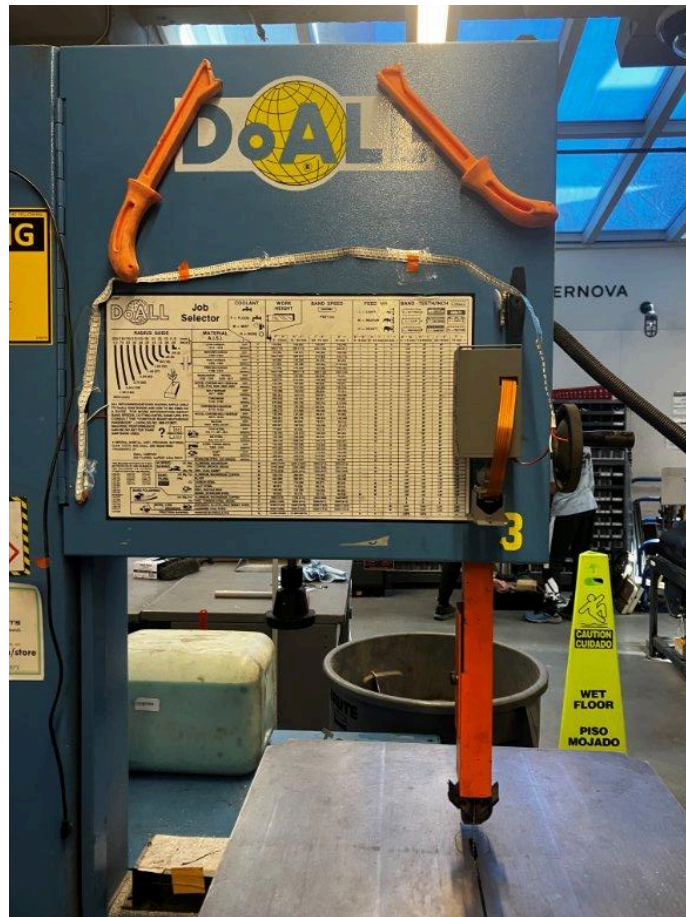


Figure 3: Bandsaw Monitor with String LEDs

Conclusion and Future Work

In conclusion, the bandsaw project demonstrates the ability of a computer vision safety system for bandsaws that enhances user awareness without interfering with machine operation. The final system successfully integrates sensing, processing, and feedback into a cohesive and functional prototype.

A future improvement could also be the addition of a pressure sensor to integrate another safety measure aside from proximity to the bandsaw blade. This could be done modularly and integrated into the current code base.

A key contribution of this work is the development of a structured and maintainable software architecture. The refactoring process significantly improved system reliability and provides a strong foundation for future development. This system also make reducing system cost as a main objective.

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